RELATIVE PSEUDO-COHERENCE DESCENDS ALONG AN FPPF COVERING

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 20639-20680, immediately following the statement at 20630-20637.

Editorial note. One cross-chapter pointer (Derived Categories of Schemes) plus six same-chapter lemmas are recorded in assembly.external_references.

Editorial note. The construction of the finite locally free pi used in the last step is displayed inside the span.

Allowed standard prerequisites (editorial).

- Flat morphisms of finite presentation are pseudo-coherent. chapter lemma lemma-flat-finite-presentation-pseudo-coherent, cited at proof lines 20641-20642. Makes each f_i pseudo-coherent so that the pullback direction follows.
- Dominating an fppf covering by a nicer one. chapter lemma lemma-dominate-fppf, cited in the proof. Replaces the covering by one whose pieces are better behaved, which is what the descent step needs.
- Pulling back and summing relative pseudo-coherence. chapter lemmas lemma-pull-relative-pseudo-coherent and lemma-summands-relative-pseudo-coherent, cited in the proof. Move the condition along the covering maps and recognise a direct summand.
- Finiteness and permanence inputs. chapter lemmas lemma-finite-morphism-relative-pseudo-coherence and lemma-permanence-pseudo-coherent with Derived Categories of Schemes, lemma perfect-lemma-cohomology-base-change, cited in the proof. Keep relative pseudo-coherence along the finite locally free morphism pi : Y -> X used at the end of the argument.

Source reference. Original lemma 05WN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13649–13649. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09UK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16995–16995. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0695: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17428–17428. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0683: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17440–17440. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09US: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17155–17155. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09UN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17070–17070. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08EU: focused-seeds/geometry/S9-morph-rel-pseudo-coherent-descends-fppf/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $X \to S$ be locally of finite type. Let $\{f_i : X_i \to X\}$ be an fppf covering of schemes. Let $E \in D_{QCoh}(\mathcal{O}_X)$. Let $m \in \mathbf{Z}$. Then E is m-pseudo-coherent relative to S if and only if each Lf_i^*E is m-pseudo-coherent relative to S.*

Proof. Assume E is m-pseudo-coherent relative to S. The morphisms f_i are pseudo-coherent by Lemma 0695. Hence Lf_i^*E is m-pseudo-coherent relative to S by Lemma 09US.

Conversely, assume that Lf_i^*E is m-pseudo-coherent relative to S for each i. Pick $S = \bigcup U_j$, $W_j \to U_j$, $W_j = \bigcup W_{j,k}$, $T_{j,k} \to W_{j,k}$, and morphisms $\alpha_{j,k} : T_{j,k} \to X_{i(j,k)}$ over S as in Lemma 05WN. Since the morphism $T_{j,K} \to S$ is flat and of finite presentation, we see that $\alpha_{j,k}$ is pseudo-coherent by Lemma 0683. Hence

$$L\alpha_{j,k}^* Lf_{i(j,k)}^* E = L(T_{i,k} \to S)^* E$$

is m-pseudo-coherent relative to S by Lemma 09US. Now we want to descend this property through the coverings $\{T_{j,k} \to W_{j,k}\}$, $W_j = \bigcup W_{j,k}$, $\{W_j \to U_j\}$, and $S = \bigcup U_j$. Since for Zariski coverings the result is true (by the definition of m-pseudo-coherence relative to S), this means we may assume we have a single surjective finite locally free morphism $\pi : Y \to X$ such that $L\pi^*E$ is pseudo-coherent relative to S. In this case $R\pi_* L\pi^* E$ is pseudo-coherent relative to S by Lemma 09UK (this is the first time we use that E has quasi-coherent cohomology sheaves). We have $R\pi_* L\pi^* E = E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \pi_* \mathcal{O}_Y$ for example by Derived Categories of Schemes, Lemma 08EU and locally on X the map $\mathcal{O}_X \to \pi_* \mathcal{O}_Y$ is the inclusion of a direct summand. Hence we conclude by Lemma 09UN. □